

GTU CSE 241 / 501 Quiz

Std Code: 142

Std Number:

2	1	0	1	0	4	0	0	4	0	1	7			
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```
#ifndef MY_HEADER
#define MY_HEADER
#include <iostream>
#include <stdexcept> // for exceptions
template<class T> // or typename , same thing.
class GtuSet{
    public :
        GtuSet() { SetElems(0) ;
                    // or SetElems.resize(0);
                }

        void add(const T& j ) {
            if( isIn(j) ) throw std::invalid_argument( ) ;
            else SetElems.push_back(j);
        }

        bool isIn( const T& j ) const {
            for(const auto& p : SetElems ) {
                if( p == j ) return true; }
            return false;
        }

        private :
            std:: vector<T>SetElems;

};

#endif
```

```

int main( ) {
    GtuSet<int> g; // could be any type.
    for (int i = 0 ; i <5 ; i++ )
        g.add( i );
    if(g.isIn(3) ) std :: cout << "set has 3" << std::endl;
    if( g.isIn( 4 ) ) { // spesific case to catch

        try { // trycatch is obligation in add function
            g.add( 4 );
        } catch( const std:: invalid_argument& e ) {
            std :: cerr <<" the error was happened " << e.what() << std:: endl;
        }
    }
}

for (int i = 2 ; i < 6 ; i++ ) {
    try { // just showing try catch again.
        g.add( i );
    } catch(const std:: invalied_argument& e ) {
        std :: cerr << " error ! " << e.what( ) << std :: endl ;
    }
}
return 0;
}

```